Technical assignment: pick and place of U-profiles with two xArm6

(drawing AMZ-SA-26-001)

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1 Task and short summary

Two UFACTORY xArm6 robots have to pick four U-profiles and place them into the hash assembly from the drawing. The second question is how the tolerances of 0.05 mm can be reached, if the robots themselves are not precise enough.

In short, what I found:

1. The pick and place itself works. In the simulation the two robots build the assembly in 16.3 s.
2. The robot alone cannot reach the tolerances. Even with ideal parts less than 1% of assemblies are in tolerance, only because of the repeatability of ±0.1 mm.
3. With parts as delivered the drawing can not be reached at all. The general tolerance allows ±0.5 mm on the length, but the dimension chain on the assembly allows only about ±0.15 mm.
4. My solution: the robots only transport the parts, a centring fixture locates and straightens them, and after joining the ends and the top surface are machined.
5. The fixture can take up to about 10 mm placement error of the robot.

Everything here is calculation and simulation, I did not have access to the real robots. The joining method is not given in the drawing. I took welding, because it is the worse case for the tolerances.

2 Check of the drawing

First I checked the masses. The cross sections are 152 mm^2 (Pos 1) and 112 mm^2 (Pos 2), with steel this gives 0.38 kg and 0.25 kg, together 1.26 kg. This is the same as in the parts list. The base plate is not in this mass and not in the parts list, so I see it as part of the fixture.

Then I went through the dimensions as chains. Main points:

Pos 1 length. $100 + 120 + 100 = 320$. If all three dimensions are ±0.05, the length of Pos 1 must be within ±0.15. By ISO 2768-m the part can be ±0.5. When the part is centred and the 120 is exact, it is even only ±0.1.

Pos 2 length. The dimension 60 is not from the axis of Pos 1 but from its outer flank. So $60 + 20 + 120 + 20 + 60 = 280$, and the width of Pos 1 (±0.3) is inside this chain. Only cutting Pos 2 to the right length does not solve it.

Orientation. 0.05 on 280 mm is 0.18 mrad, about 0.01°. For the robot this is the hardest point, because a small angle error of the joints is multiplied by the part length.

Straightness. A parallelism of 0.05 on a flank also means the flank must be straight within 0.05. Cold rolled profiles can be bent by about 0.17% of the length (value from a secondary source, the EN 10162 text was not available to me). For 320 mm this is around 0.5 mm.

Contact. Both profiles are open to the bottom. So Pos 2 stands on the ends of its legs, these are 8 small strips of 2×40 mm. A gap of 0.05 on such sawn edges is not possible without machining.

Not defined in the drawing: the joining method, the datums for the geometric tolerances, and the base plate.

3 Tolerance calculation (Monte Carlo)

To see which process steps are needed, I wrote a Monte Carlo model of the assembly in Python. For every assembly the dimensions, bow, twist, heights and placement errors are taken randomly, and then all 8 requirements of the drawing are checked. For every process variant 200 000 assemblies were calculated. Input values are in Table 1.

Table 1: Input values for the tolerance model

Parameter	Value	Where from
Robot repeatability	±0.1 mm (as 3σ)	xArm6 datasheet
Part position in gripper	±0.2 mm	own estimate
Length, width, height	ISO 2768-m	drawing
Bow	up to 0.17% of length	secondary source to EN 10162
Twist	up to 1°/m	own estimate
Step between leg ends	up to 0.2 mm	own estimate
Fixture locators	±0.01 mm	ground locators
Milling after joining	±0.02 mm	one clamping

The result is in Fig. 1. Every row is one process variant, every next row adds one step.

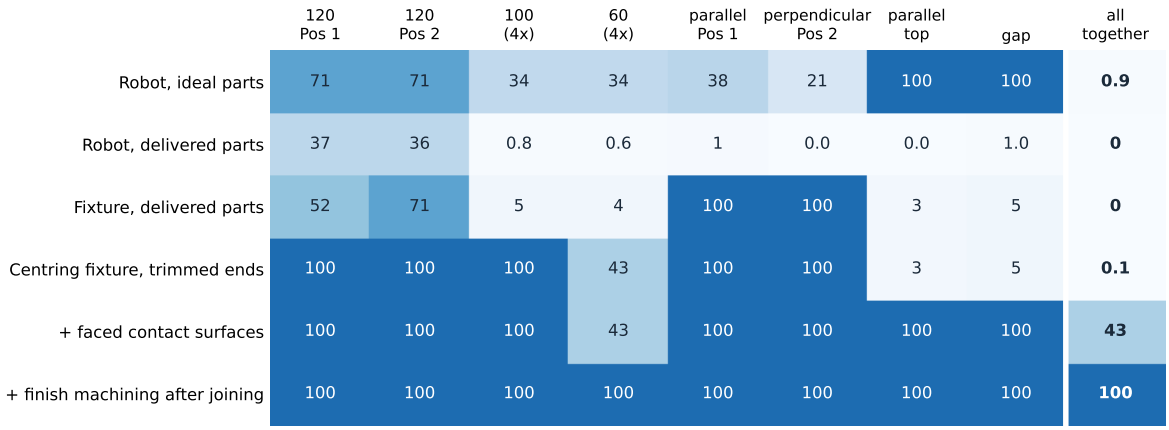


Figure 1: Percentage of assemblies that meet each requirement (normal distribution of part errors).

What can be seen from Fig. 1:

- Robot only: even with ideal parts the orientation and the dimensions 100 and 60 fail. This is the answer to the reality check question, the precision has to come from somewhere else.

- With a fixture the position is good, but still almost no assembly is OK. The reason is the height of the profiles, the leg ends and the length.
- Machining the contact surfaces solves height and gap. The dimension 60 is only OK when the ends are machined after joining, measured from the real welded frame.

The 100% in the last row comes only from the assumed milling accuracy, welding distortion is not inside. So it means "no geometric conflict is left", not a real yield.

I also tried a uniform distribution instead of normal. Then the variant with machined contact surfaces goes down from 43% to 9%, but the last row stays at 100%. So with machining after joining the result does not depend much on the supplier.

4 How to reach the tolerances

The idea is to give each accuracy level to the part of the system that can do it:

Level	What it does	Accuracy
Robots	bring the part into the capture zone	ca. 1 mm
Fixture	centre, straighten and hold during joining	ca. 0.01 mm
Machining and measuring	final reference surfaces, check	ca. 0.02 mm

4.1 Fixture

The fixture has stops on both sides of each profile which close together (centring), so the profile is located on its middle plane. With a stop only on one side, the width tolerance would move the axis. The concept is in Fig. 2, the STEP file is in the repository.

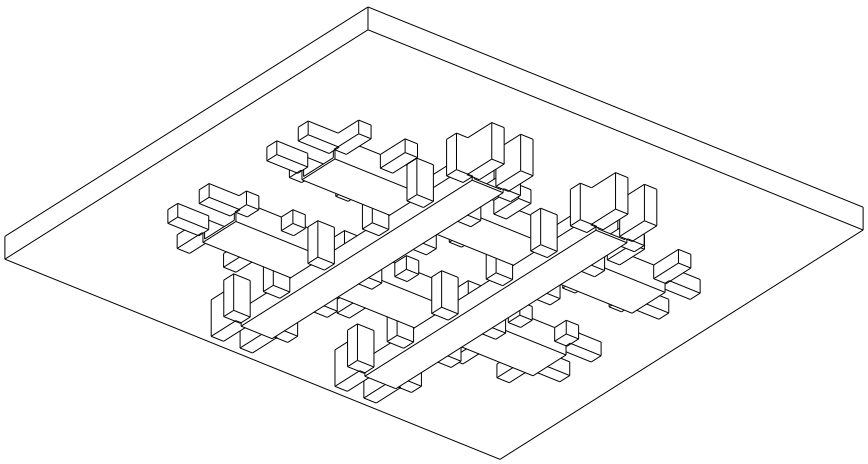


Figure 2: Fixture concept. 7 stops on each Pos 1 flank, 6 on each Pos 2 flank, end stops on the legs.

Some points for the fixture design:

- To straighten a bow of 0.5 mm a force of about 300 N is needed (three point bending, $F = 48EI\delta/L^3$, $I \approx 3.7 \cdot 10^4$ mm^4 for Pos 1). The xArm6 payload is 5 kg, so the robot can not do this, the clamps have to.
- The bow left between two stops goes with $(s/L)^2$. Under Pos 2 there is Pos 1, so the stops can not be placed evenly. If they are put directly next to the Pos 1 flanks, the biggest span

is 66 mm. With even spacing it would be 105 mm, and then only 48% of the assemblies are OK in perpendicularity.

- The stress at 300 N is only about 13 MPa, much below the yield strength. So the profile springs back after unclamping, and stays straight only as much as the welds hold it.
- The end stops must touch the legs. A stop in the middle of the open profile goes under the web and does not touch the part.
- Lead-in: 4 mm clearance and 7 mm chamfer on each side. The chamfer must not stick out over the stop face, otherwise it lifts the part when clamping.

4.2 Process steps

1. Incoming check of the parts: length, width, bow.
2. Machine the leg ends of Pos 2 and the top of Pos 1, leave material on the ends.
3. The robots put the parts into the fixture. First the end stops close, then the side stops, then the top clamps.
4. Welding with low and symmetric heat input. With 12 μm/(m K) already 13 K temperature difference gives 0.05 mm on 320 mm. That is why the fixture should be from steel and not aluminium.
5. Machining of the ends and the top of Pos 2 in one clamping, from the welded frame.
6. Measuring, and correcting the clamping and welding sequence with the results.

If machining after welding is not allowed, another way is selective assembly: measure the width of each Pos 1 and cut each Pos 2 to the matching length.

5 Robot cell and simulation

The cell was simulated in MuJoCo, with the xArm6 geometry from the official UFACTORY ROS2 package. The two robots stand opposite each other, 800 mm apart, and each has its own part stacks (Fig. 3). Each robot places one Pos 1 and then one Pos 2. Pos 2 can only be placed when both Pos 1 are in the fixture. As gripper I use a tool with two suction cups on the closed web. A magnetic gripper would also be possible, because S235 is magnetic.

At a distance of 120 mm the two wrists do not fit next to each other above the middle of the parts. So each robot grips the part 80 mm out of the middle, to its own side.

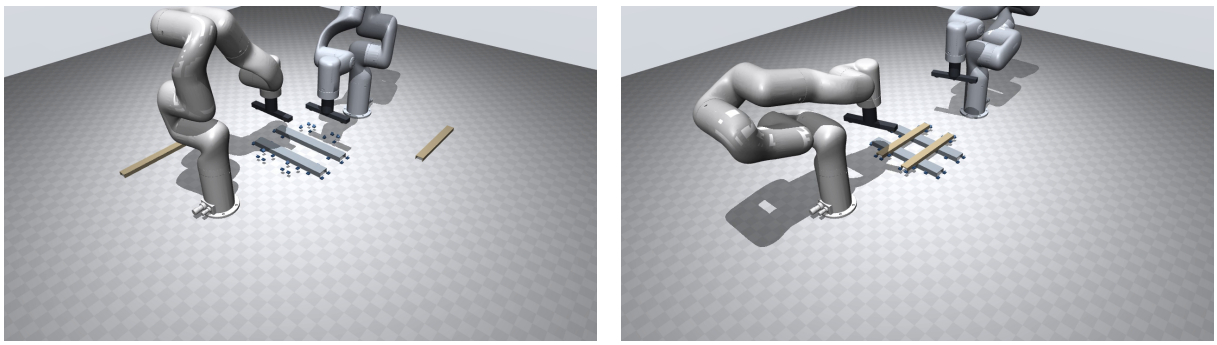


Figure 3: Simulation: placing of the lower level (left) and the finished assembly (right).

5.1 Capture range of the fixture

To check how much robot error the fixture can take, the parts were dropped from 12 mm above the seat with a random error up to 16 mm and ±1°, 600 times per part type. Then the jaws were closed. A part is counted as seated if it is within 0.1 mm and 1 mrad in the plane and 0.3 mm in height.

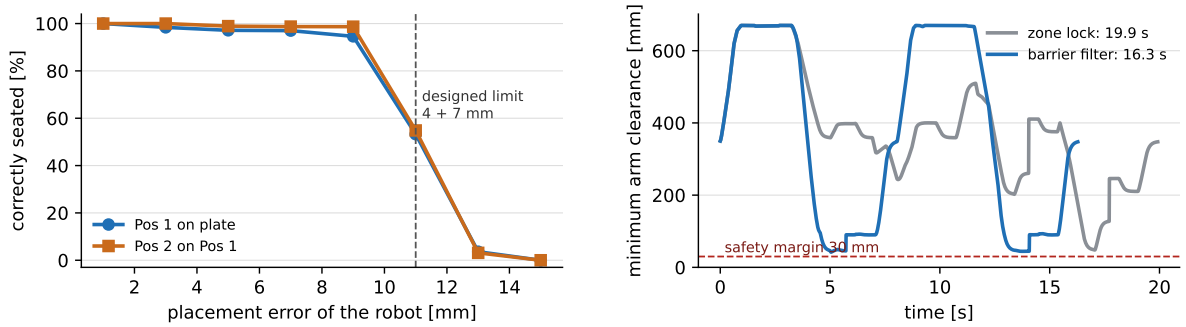


Figure 4: Left: capture curve (60–90 tries per point). Right: smallest distance between the robots during one cycle.

As can be seen in Fig. 4 (left), up to 10 mm error 95–100% of the parts are seated. At 11 mm it drops, this is exactly clearance plus chamfer. The parts stay about 0.14 mm too high because of the friction in the jaws, so top clamps are necessary. The simulation uses rigid contact, so it shows the capture geometry, but it says nothing about accuracy in micrometres.

5.2 Two robots in one workspace

The robots can collide above the fixture. The simple solution is a zone lock: only one robot is allowed in the common zone. As a second variant both robots move freely and a control barrier function (CBF) filter is added. h is the distance between capsules around the robot links minus a safety margin of 30 mm. The joint velocity is changed as little as possible:

$$\dot{q}^{*} = \arg\min_{\dot{q}} \|\dot{q} - \dot{q}_{\mathrm{nom}}\|_{W}^{2} \quad \text{s.t.} \quad \dot{h}(q,\dot{q}) \geq -\alpha h$$

From this $h(t) \geq h(0)e^{-\alpha t}$, so if the robots start in a safe state they stay safe. The QP is solved at 100 Hz with Hildreth's method.

	Zone lock	CBF filter
Cycle time	19.9 s	16.3 s
Smallest distance between robots	47.8 mm	42.1 mm
Filter active	–	3% of the steps

The cycle is 18% shorter. This comes mainly from the parallel work, the filter is active only in 3% of the steps, it is more a safety net. With gripping in the middle of the parts the filter stopped both robots at the 30 mm margin. So it avoids the collision, but it can not solve the deadlock, the task plan itself must be feasible. Also the guarantee is only valid if the robot follows the velocity command, and it is not a replacement for a certified safety function.

6 Calibration and placing on the real robot

Calibration. For the cell the transformation between the two robot bases is needed. One touch on a plane with normal n in point p gives one row $[n^{\top}, (p \times n)^{\top}]$ of the measurement Jacobian.

If only the plate is touched, the rank is 3, the shift in the plane and the rotation around the vertical axis can not be found. With 3-2-1 touches on the plate, a long side and an end face the rank is 6. So the datum faces of the fixture are also needed for the calibration.

Placing. In the simulation the part is dropped above the seat. On the real robot I would put it down with admittance control, $M\ddot{e} + D\dot{e} + Ke = f$. With $V = \frac{1}{2}\dot{e}^\top M\dot{e} + \frac{1}{2}e^\top Ke$ follows $\dot{V} = -\dot{e}^\top D\dot{e} + \dot{e}^\top f$, so the robot is passive in contact and stays stable with the passive fixture.

7 Limitations and next steps

- The motion in the simulation is kinematic, with position controlled joints and without payload dynamics.
- Twist, step of the leg ends and position in the gripper are estimated. The order of the process variants stays the same also for other values.
- Welding distortion and springback after unclamping are not in the model. This is the biggest open risk.
- In the simulation the fixture has only two fingers per side, the tolerance model and the CAD have the full stop layout.

Next steps on hardware: measure the repeatability with a dial gauge, do the capture test with 3D printed fixture blocks, measure the bow on some real parts, and weld test frames to see the distortion.

Code, CAD and videos (also a side by side video of both variants): [repository link].